

ROBERT VOKÁČ, BSc

C++-focused Software Engineer - Cross-Platform Systems & Compatibility Layers

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PROFILE

Software Engineer with long-term production experience and substantial independent work in **modern C++ systems programming**.

I build **cross-platform C++ frameworks**, platform abstraction layers, and compatibility systems over **SDL3**, **OpenGL**, and related technologies. My work focuses on portable architecture, legacy API behavior, rendering/input/audio abstraction, and application portability across Windows, Linux, WebAssembly, and Android.

Seeking C++ roles in systems programming, cross-platform development, framework engineering, tooling, or platform abstraction.

RELEVANT C++ ENGINEERING EXPERIENCE

Independent C++ Systems Developer – OpenEggbert

Open-source engineering work

2024 – present

Designed and built modern C++ framework and compatibility-layer projects focused on portability, API behavior reproduction, and platform abstraction.

Main technologies: **C++20/23**, **SDL3**, **OpenGL**, **bgfx**, **CMake**, **Windows**, **Linux**, **WebAssembly**, **Android**

CNA - Cross-Platform C++ Framework

GitHub: <https://github.com/openeggbert/cna>

Custom C++ implementation of an XNA-style API over SDL3, validated by a working C# XNA game port running natively on desktop, Android, and WebAssembly.

- Designed a cross-platform C++ framework layer over SDL3
- Built application loop, input, rendering, audio, and resource systems
- Added rendering backends for SDL_Renderer, OpenGL, and bgfx
- Ported a C# / XNA application to native C++
- Replaced .NET, MonoGame, and original XNA dependencies with native C++/SDL3 code
- Produced stable builds from one codebase for Windows, Linux, WebAssembly, and Android

FreeDirect + FreeAPI - Legacy Graphics & System Compatibility Layer

GitHub: <https://github.com/openeggbert/free-direct>

GitHub: <https://github.com/openeggbert/free-api>

Modern C++ compatibility layer providing DirectDraw-style rendering and selected WinAPI-style behavior over SDL3.

- Created DirectDraw-style 2D rendering abstractions
- Built event loop, window/input handling, and platform integration
- Enabled legacy Windows applications to run through a modern SDL3-based backend
- Reproduced selected DirectX 3 / DirectDraw and WinAPI-style behavior without obsolete SDK dependencies
- Continued work on broader compatibility for legacy Windows applications

Nova3D – Experimental 3D Runtime Layer

GitHub: <https://github.com/openeggbert/nova-3d>

Experimental higher-level 3D runtime layer built on top of CNA.

- Designed an Urho3D-inspired API layer over a custom C++ runtime stack
- Implemented basic scene structure, camera handling, mesh rendering, and demo logic
- Demonstrates growth from 2D runtime compatibility toward 3D engine architecture

Hive - Modular Backend Platform

GitHub: <https://github.com/robertvokac/hive>

Archived 40k+ LOC modular backend platform demonstrating architecture, metadata-driven design, and larger-codebase experience.

- Designed metadata-driven backend architecture
- Built custom ORM concepts, scheduler, triggers, validation pipeline, and REST generation
- Created plugin-oriented structure allowing extension without modifying core code

PRODUCTION SOFTWARE EXPERIENCE

Software Engineer - Assist

2017 – present

Production software engineering role focused on backend systems, SQL, legacy codebases, and long-term maintenance.

- Maintained and improved production backend systems using Java and SQL
- Diagnosed production issues in large legacy systems
- Worked with long-term architectural constraints and stability requirements
- Improved reliability and maintainability of existing backend software

Technologies: Java, SQL, Git, Bash

CORE SKILLS

Languages: C++20/23, SQL, Java

C++ / Systems: STL, RAII, smart pointers, resource ownership, modular architecture

Cross-platform: Windows, Linux, WebAssembly, Android

Graphics / Platform: SDL3, OpenGL, bgfx, DirectDraw-style rendering, platform abstraction

Build / Tools: CMake, Git, Bash, CLion, Visual Studio

Backend / Data: SQL, SQLite, REST API design, ORM concepts

Core areas: compatibility layers, legacy behavior reproduction, rendering/input/audio systems

EDUCATION

BSc in Computer Science

Czech University of Life Sciences Prague

2014 – 2017

LANGUAGES

Czech: Native

English: B2